SATGURU STUDY CENTRE
Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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SATGURU STUDY CENTRE
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CLASS : 10

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COMMON QUARTERLY EXAMINATION-2024-25

Time Allowed : 3.00 Hours] **MATHEMATICS** [Max. Marks : 100

SECTION – I

I. Answer all of the following: **14x1=14**

1. If the ordered pairs (a+2; 4) and (5, 2a+b) are equal then (a,b) is
   a) (2,-2) b) (5,1) c) (2,3) d) (3,-2)
2. $f(x) = (x+1)^3 - (x-1)^3$ represents a function which is ----------
   a) Linear b) Cubic c) Reciprocal d) Quadratic
3. $7^{4k} \equiv$ -------------- (mod 100)
   a) 1 b) 2 c) 3 d) 4
4. The next term of the sequence $^3/_{16}, {}^1/_8, {}^1/_{12}, {}^1/_{18}$, ... is
   a) $^1/_{24}$ b) $^1/_{27}$ c) $^2/_3$ d) $^1/_{81}$
5. Euclid division algorithm is a repeated application of division lemma until we get remainder as ---------
   a) 1 b) 3 c) 0 d) 2
6. The solution of $(2x - 1)^2 = 9$ is equal to
   a) -1 b) 2 c) -1, 2 d) None of these
7. $\dfrac{3y-3}{y} \div \dfrac{7y-7}{3y^2}$ is
   a) $\dfrac{9y}{7}$ b) $\dfrac{9y^3}{21y-21}$ c) $\dfrac{21y^2-42y+21}{3y^2}$ d) $\dfrac{7(y^2-2y+1)}{y^2}$
8. If in triangle ABC and DEF, $\dfrac{AB}{DE} = \dfrac{BC}{FD}$, then they will be similar, when
   a) $\angle B = \angle E$ b) $\angle A = \angle D$ c) $\angle B = \angle D$ d) $\angle A = \angle F$
9. If in ΔABC, DE||BC, AB = 3.6 cm, AC = 2.4 cm and AD = 2.1cm then the length of AE is
   a) 1.4 cm b) 1.8 cm c) 1.2 cm d) 1.05 cm
10. The slope of the line joining (12,3), (4,a) is $^1/_8$. The value of 'a' is
    a) 1 b) 4 c) -5 d) 2
11. (2, 1) is the point of intersection of two lines
    a) $x - y - 3 = 0; 3x - y - 7 = 0$ b) $x + y = 3; 3x + y = 7$
    c) $3x + y = 3; x + y = 7$ d) $x + 3y - 3 = 0; x - y - 7 = 0$.
12. $\tan\theta \operatorname{cosec}^2\theta - \tan\theta$ equal to
    a) $\sec\theta$ b) $\cot^2\theta$ c) $\sin\theta$ d) $\cot\theta$
13. If $\sin\theta = \cos\theta$ then $2\tan^2\theta + \sin^2\theta - 1$ is equal to
    a) $-^3/_2$ b) $^3/_2$ c) $^2/_3$ d) $-^2/_3$
14. f = {(2,a), (3,b), (4,b), (5,c)} is a
    a) Identity function b) One-one function
    c) Many-one function d) Constant function

SECTION - II

II. Answer any 10 questions. Question No. 28 is compulsory. **10x2=20**

15. Let A = {1, 2, 3} and B = {x/x is a prime number less than 10}. Find AxB and BxA.
16. If A = {-2, -1, 0, 1, 2} and f: A→B is an onto function defined by $f(x) = x^2+x+1$ then find B.
17. Find the first four terms of $a_n = n^3-2$.
18. Find the sum $3 + 1 + {}^1/_3 + \ldots\ldots\ldots \infty$
19. Find the 12^{th} term from the last term of the A.P -2, -4, -6, ..., -100.
20. Find the LCM of $8x^4y^2$, $48x^2y^4$.

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21. Determine the quadratic equation whose sum and product of -9 and 20.

22. Check whether AD is bisector of $\angle A$ of ΔABC in the following.
AB = 5cm, AC = 10cm, BD = 1.5cm and CD = 3.5 cm.

23. What is the inclination of a line whose slope is '0'.

24. Show that the straight lines $2x+3y-8=0$ and $4x+6y+18=0$ are parallel.

25. Prove that $\sec\theta - \cos\theta = \tan\theta \sin\theta$.

26. Prove that $\sqrt{\dfrac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta$

27. Show that the points P(-1.5,3), Q(6,-2), R(-3,4) are collinear.

28. $P = \dfrac{x}{x+y}$, $Q = \dfrac{y}{x+y}$ then find $\dfrac{1}{P^2-Q^2}$

SECTION - III

III. Answer the following any 10 questions. Q.No.42 is compulsory. **10x5=50**

29. If the function f is defined by $f(x) = \begin{cases} x+2 & \text{if } x>1 \\ 2 & \text{if } -1\le x\le 1 \\ x-1 & \text{if } -3<x<-1 \end{cases}$ Find the values of

i) f(3) ii) f(0) iii) f(-1.5) iv) f(2) + f(-2)

30. Let $f(x) = x^2-1$. Find (i) fof (ii) fofof

31. Find the HCF of 396, 504,, 636.

32. Rekha has 15 square colour papers of sizes 10 cm, 11 cm, 12 cm.....24cm. How much area can be decorated with these colour papers?

33. Solve $x + 2y - z = 5$, $x - y + z = -2$, $-5x - 4y + z = -11$.

34. Find the GCD of $6x^3 - 30x^2 + 60x - 48$ and $3x^3 - 12x^2 + 21x - 18$.

35. If $4x^4 - 12x^3 + 37x^2 + bx + a$ is a perfect square, find the values of 'a' and 'b'.

36. State and prove Thales Theorem.

37. Find the value of K, if the area of a quadrilateral is 28 sq. units, whose vertices are (-4, -2), (-3, k), (3, -2) and (2, 3).

38. Without using pythagoras theorem, show that the points (1, -4), (2, -3) and (4, -7) form a right angled triangle.

39. Find the equation of a straight line passing through (5, -3) and (7, -4).

40. If $\sqrt{3}\sin\theta - \cos\theta = 0$ then show that $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

41. Given A = {1, 2, 3}, B = {2, 3, 5} , C = {3, 4} and D = {1, 3, 5}, check if $(A\cap C) \times (B\cap D) = (A\times B) \cap (C\times D)$ is true?

42. Find the sum of all natural numbers between 100 and 10,000 which are divisible by 11.

SECTION - IV

IV. Answer the following. **2x8=16**

43. a) Construct a triangle similar to a given triangle PQR with its sides equal to $^3/_5$ of the corresponding sides of the ΔPQR (scale factor $^3/_5 < 1$)

(OR)

b) Construct a ΔABC such that AB = 5.5 cm, $\angle C = 25^0$ and the altitude from C to AB is 4.5 cm.

44. a) Graph the following linear function $y = {^1/_2}x$. Identify the constant of variation and verify it with the graph. Also (i) Find y when x=9, (ii) Find x when y = 7.5.

(OR)

b) A bus is travelling at a uniform speed of 50 km/hr. Draw the distance - time graph and hence find i) the constant of variations ii) how far will it travel in 90 minutes? iii) the time required to cover a distance of 300 km from the graph.

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